

- There are constraints to evolution: Natural selection can act only on available variation; structures result from modified ancestral anatomy; adaptations are often compromises; and chance, natural selection, and the environment interact.

? How might secondary sex characteristics in males differ from those in females in a species in which females compete for mates?

TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

- Natural selection changes allele frequencies because some _____ survive and reproduce better than others.
(A) alleles (C) species
(B) loci (D) individuals
- No two people are genetically identical, except for identical twins. The main source of genetic variation among humans is
(A) new mutations that occurred in the preceding generation.
(B) genetic drift.
(C) the reshuffling of alleles in sexual reproduction.
(D) environmental effects.

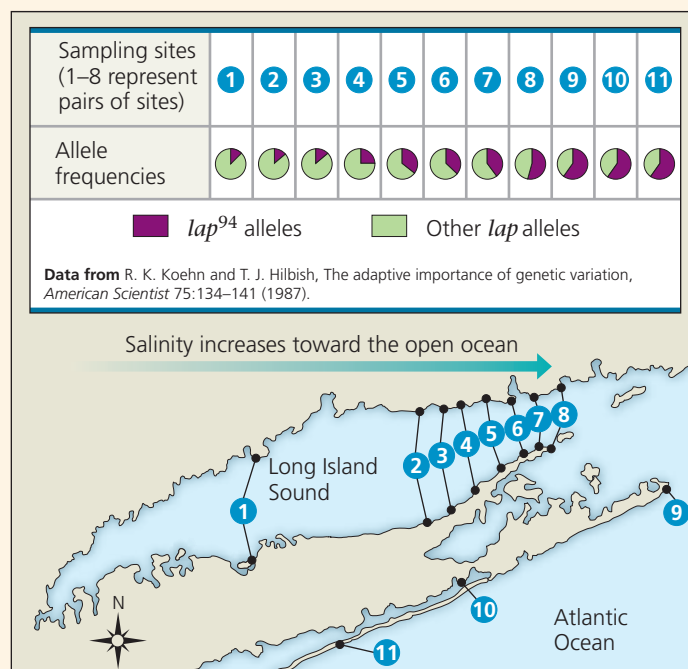
Levels 3-4: Applying/Analyzing

- If the nucleotide variability of a locus equals 0%, what is the gene variability and number of alleles at that locus?
(A) gene variability = 0%; number of alleles = 0
(B) gene variability = 0%; number of alleles = 1
(C) gene variability = 0%; number of alleles = 2
(D) gene variability > 0%; number of alleles = 2
- One population of a species has 25 individuals, all with genotype *AA*; a second population of this species has 40 individuals, all with genotype *aa*. Assume that these populations live far apart but in similar environmental conditions. Based on this information, the observed genetic variation most likely resulted from
(A) genetic drift. (C) nonrandom mating.
(B) gene flow. (D) directional selection.
- A fruit fly population has a gene with two alleles, *A1* and *A2*. Tests show that 70% of the gametes produced in the population contain the *A1* allele. If the population is in Hardy-Weinberg equilibrium, what proportion of the flies carry both *A1* and *A2*?
(A) 0.7 (B) 0.49 (C) 0.42 (D) 0.21

Levels 5-6: Evaluating/Creating

- EVOLUTION CONNECTION** Using at least two examples, explain how the process of evolution is revealed by the imperfections of living organisms.
- SCIENTIFIC INQUIRY • INTERPRET THE DATA** For populations of the marine mussel *Mytilus edulis* around Long Island, New York, researchers measured the frequency of a particular allele (*lap*⁹⁴) for an enzyme involved in regulating the mussel's saltwater balance. They presented their data as a series of pie charts linked to sampling sites within Long Island Sound, where the salinity is highly variable, and along the coast of the open ocean, where salinity is constant. (a) Create a data table for the 11 sampling sites by estimating the frequency of *lap*⁹⁴ from the pie charts. (b) Graph the frequencies for sites 1–8 to show how the frequency of this allele changes with increasing salinity in Long Island Sound (from southwest to northeast). Evaluate how the data from sites 9–11 compare with the data from the sites within the Sound. (c) Considering the various mechanisms that can alter allele frequency, construct a hypothesis that explains the patterns

in the data and that accounts for the following observations: (1) The *lap*⁹⁴ allele helps mussels maintain osmotic balance in water with a high salt concentration but is costly to use in less salty water; and (2) mussels produce larvae that can disperse long distances before they settle on rocks and grow into adults.



- WRITE ABOUT A THEME: ORGANIZATION** Heterozygotes at the sickle-cell locus produce both normal and abnormal (sickle-cell) hemoglobin (see Concept 14.4). When hemoglobin molecules are packed into a heterozygote's red blood cells, some cells receive relatively large quantities of abnormal hemoglobin, making these cells prone to sickling. In a short essay (approximately 100–150 words), explain how these molecular and cellular events lead to emergent properties at the individual and population levels of biological organization.

9. SYNTHESIZE YOUR KNOWLEDGE



The lake shown in this photograph, called a kettle lake, formed 14,000 years ago when a glacier that covered the surrounding area melted. Initially devoid of animal life, over time the lake was colonized by invertebrates and other animals. Hypothesize how mutation, natural selection, genetic drift, and gene flow may have affected populations that colonized the lake.

For selected answers, see Appendix A.

Explore Scientific Papers with Science in the Classroom AAAS

How does a native species evolve in response to an introduced species? Go to "There's a New Kid in Town" at www.scienceintheclassroom.org.

➔ **Instructors:** Questions can be assigned in Mastering Biology.